Package 'AnnotatoR'

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Title An R package for the creation of general purpose, IonReporter-compatible variant annotation files.

Version 0.1.0

Authors Dionysios Fanidis [aut, cre] https://orcid.org/0000-0002-4053-2090

Description Given a set of HGNC gene symbols, the package retrieves on demand known variants and respective annotation from publicly available databases/predictors. Although formatted as IonRerorter-compatible files, results can be used for any other purpose.

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Encoding UTF-8

Imports EnsDb.Hsapiens.v86,
EnsDb.Hsapiens.v75,
BSgenome.Hsapiens.UCSC.hg38,
ensembldb,
GenomicRanges,
HGNChelper,
httr,
jsonlite,
IRanges,
progressr,
rjson,
Rsamtools,
S4Vectors,
stringr,
utils,
SummarizedExperiment,
VariantAnnotation,
liftOver,
dplyr

Config/roxygen2/version 8.0.0

Contents

`annotate` *Create general purpose, IonReporter-compatible variant annotation files.*

Description

A wrapper function for downloading and organizing annotation files.

Usage

```
annotate(
  gns,
 annotators = c("revel", "alphamissense", "clinvar_sig", "gnomad_af", "intervar",
    "mutation_taster", "all"),
  databases = c("gnomad_man", "gnomad_auto", "clinvar", "lovd3", "all"),
  panelName = "./results/panel",
  path = "./dbs",
  type = c("exomes", "genomes", "both"),
  liftover = FALSE,
  saveRaw = FALSE
)
```

Arguments

`gns`	Gene names (HGNC) character vector.
`annotators`	Annotator resources to retrieve data from.
`databases`	Databases to retrieve variants from. The argument is taken into consideration when ‘intervar‘, ‘gnomad_af‘, ‘mutation_taster‘ or ‘all‘ annotators are selected. ‘gnomad_man‘ expects manually downloaded gnomAD csv files placed in ‘gnomad/manual‘ subdirectory of the directory set in ‘path‘ argument (see below).
`panelName`	Path and prefix for output file.
`path`	Path to download directory.
`type`	Type of gnomAD data to download. Required when ‘annotators‘ argument is set to/includes ‘gnomad_af‘, ‘mutation_taster‘, ‘intervar‘, ‘all‘
`liftover`	Should variants be lifted over to hg19? (Default hg38)
`saveRaw`	Save all variants without any annotation. Applicable only for ‘intervar‘ and ‘all‘ annotators.

Value

VCF-like annotation annotation files compatible with IonReporter.

`annotateInterVar` *Annotate via InterVar’s API.*

Description

Predict variants’ pathogenicity according to ACMG criteria as implemented by InterVar.

Usage

```
annotateInterVar(vcf, liftover)
```

Arguments

`vcf`	A variant table as the one created by ‘collectVars‘
`liftover`	Were variants lifted over?

Value

A nested list with InterVar’s ACMG and other annotation pieces per variant.

clinvar_fetch *Fetch ClinVar variants.*

Description

Collect known variants for specific genes from ClinVar

Usage

```
clinvar_fetch(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar variants.

collectVars *Collect known variants.*

Description

A wrapper function to collect known variants for specific genes from public resources (gnomAD, ClinVar, LOVD3).

Usage

```
collectVars(
  gns,
  databases = c("gnomad_man", "gnomad_auto", "clinvar", "lovd3", "all"),
  path,
  type = c("exomes", "genomes", "both"),
  liftover
)
```

Arguments

gns	Gene names (HGNC) character vector.
databases	Databases to retrieve variants from.
path	Path to download directory.
type	Type of gnomAD data to download. Used when ‘gnomad_auto‘ database is selected.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with gene variants from the selected databases.

`constructAF` *Retrieve gnomAD allele frequencies.*

Description

A wrapper function to retrieve population allele frequencies as recorded in gnomAD. ‘gnomad_man‘ searches for manually downloaded gnomAD csv files in ‘gnomad/manual‘ inside the ‘dbs‘ directory. If not found, it downloads (upon approval) gnomAD raw files instead.

Usage

```
constructAF(
  gns,
  path,
  databases = c("gnomad_man", "gnomad_auto"),
  type = c("exomes", "genomes", "both"),
  liftover
)
```

Arguments

`gns`	Gene names (HGNC) character vector.
`path`	Path to download directory.
`type`	Type of gnomAD data to download. Used when ‘gnomad_auto‘ database is selected.
`liftover`	Should variants be lifted over?

Value

A VCF-like data.frame with gnomAD allele frequency in the 3rd column (‘ID‘).

`constructAM` *Contruct an AlphaMissense annotation file.*

Description

Download and organize AlphaMissense predictions for genes of interest.

Usage

```
constructAM(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with AlphaMissense pathogenicity prediction in the 3rd column (‘ID‘).

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constructClinVar *Retrieve ClinVar significance.*

Description

A wrapper function to retrieve ClinVar variant significance from gnomAD. Manually downloaded gnomAD files are required.

Usage

```
constructClinVar(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar germline classification values in the 3rd column (‘ID‘).

---

constructClinVar_deprecated *Contruct ClinVar annotation file.*

Description

Download and organize ClinVar significance classification with review status level for specific genes.

Usage

```
constructClinVar_deprecated(gns, path, liftover)
```

Arguments

`gns`	Gene names (HGNC) character vector.
`path`	Path to download directory.
`liftover`	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar germline classification values and number of review stars in the 3rd column (‘ID‘).

`constructREVEL` *Contruct REVEL annotation file.*

Description

Download and organize REVEL meta-predictions for specific genes.

Usage

```
constructREVEL(gns, path, liftover)
```

Arguments

`gns`	Gene names (HGNC) character vector.
`path`	Path to download directory.
`liftover`	Should variants be lifted over?

Value

A VCF-like data.frame with REVEL deleteriousness prediction score in the 3rd column (‘ID‘).

`download_am` *Download AlphaMissense tsv files*

Description

This function downloads AlphaMissense tsv file.

Usage

```
download_am(path)
```

Arguments

`path`	Where to download data

Value

AlphaMissense vcf files

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download_clinvar *Download ClinVar files*

Description

This function downloads ClinVar vcf file.

Usage

```
download_clinvar(path)
```

Arguments

path	Where to download data

Value

ClinVar vcf file.

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download_gnomad *Global functions Download gnomad chromosome files*

Description

Downloads gnomAD vcf files according to specified chromosome.

Usage

```
download_gnomad(chr, type = c("exomes", "genomes", "both"), path)
```

Arguments

chr	Chromosome name
type	Type of gnomAD data to download.
path	Where to download data

Value

gnomAD vcf files

`download_lovd3` *Download LOVD3 gene variants.*

Description

This function downloads LOVD3 gene.

Usage

```
download_lovd3(gns, path)
```

Arguments

`gns`	Gene names
`path`	Where to download data

Value

LOVD3 gene files.

`download_revel` *Download REVEL chromosome files.*

Description

This function downloads REVEL vcf files per chromosome.

Usage

```
download_revel(chr, path)
```

Arguments

`chr`	Chromosome name
`path`	Where to download data

Value

REVEL vcf files.

`gnomad_fetch_auto` *Fetch gnomAD variants.*

Description

Collect known variants for specific genes from gnomAD.

Usage

```
gnomad_fetch_auto(
  map,
  type = c("exomes", "genomes", "both"),
  path,
  liftover,
  af = FALSE
)
```

Arguments

`map`	Gene data as in map.
`type`	Type of gnomAD data to download. Used when 'gnomad_auto' database is selected.
`path`	Path to download directory.
`liftover`	Should variants be lifted over?
`af`	Maintain allele frequencies?

Value

A VCF-like data.frame with gnomAD variants.

`gnomad_fetch_manual` *Fetch manually downloaded gnomAD variants.*

Description

Collect known variants for specific genes downloaded manually from gnomAD

Usage

```
gnomad_fetch_manual(map, path, liftover, af = FALSE, sig = FALSE)
```

Arguments

`map`	Gene data data.frame as returned from 'map_fetch'.
`path`	Path to download directory.
`liftover`	Should variants be lifted over?
`af`	Maintain allele frequencies?

Value

A VCF-like data.frame with gnomAD allele frequencies (af = TRUE) or ClinVar significance values (sig = TRUE) in the 3rd column ('ID').

`lift` — *LiftOver hg38 to hg19 genomic coordinates using rtracklayer functionalities*

Description

LiftOver hg38 to hg19 genomic coordinates using rtracklayer functionalities

Usage

```
lift(vcf, ...)
```

Arguments

`vcf`	A vcf dataframe with CHR and POS columns
`...`	Arguments for methods

Value

A GRanges object of lifted over genomic coordinates

`lovd3_fetch` — *Fetch LOVD3 variants.*

Description

Collect known variants for specific genes from LOVD3

Usage

```
lovd3_fetch(gns, path, liftover)
```

Arguments

`gns`	Gene names (HGNC) character vector.
`path`	Path to download directory.
`liftover`	Should variants be lifted over?

Value

A VCF-like data.frame with LOVD3 variants.

`map_fetch` *Annotate genes using Ensembl data*

Description

Annotate genes using Ensembl data

Usage

```
map_fetch(ensdb, gns, trans = FALSE)
```

Arguments

`ensdb`	An EnsDb package.
`gns`	Gene names (HGNC) character vector.
`trans`	Should transcript level data be returned?

Value

A data frame with genes (default) or ensembl transcripts as rows and Ensembl annotation as columns.

`process_json` *Handling errors*

Description

Handling errors

Usage

```
process_json(j)
```

Arguments

`j`	json file

Value

LOVD3 gene files.

`revstatus_map` *Review status on aggregate records (VCV and RCV) - ClinVar*

Description

Review status on aggregate records (VCV and RCV) - ClinVar

Usage

```
revstatus_map
```

`taste` *Annotate via MutationTaster's API.*

Description

Retrieve variants' pathogenicity predictions from MutationTaster.

Usage

```
taste(vcf, liftover)
```

Arguments

`vcf`	A variant table as the one created by 'collectVars'.
`liftover`	Whether variants have been lifted over (boolean).

Value

Annotated variants with MutationTaster2021 and MutationTaster2025 predictions for GRCh37 (liftover = TRUE) and GRCh38 (liftover = FALSE) variants, respectively.

`vcf_header_allele` *VCF headers for IonReporter*

Description

VCF headers for IonReporter

Usage

```
vcf_header_allele
```

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